

Amanda Guerra

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EDUCATION

The University of Texas Rio Grande Valley
Bachelor of Science in Electrical Engineering (Junior) **GPA: 3.73**

Expected Graduation May 2028

Relevant Coursework: Electronics, Digital Systems, Signals & Systems, Automatic Control, Microprocessors

SKILLS

- | | | |
|-------------------------------|-------------------------------|----------------------|
| • MATLAB | • Microsoft Office | • Circuit Design |
| • C / C++ | • Altium Designer | • Soldering |
| • Autodesk Inventor | • KiCad | • Python |
| • Autodesk Revit | • Arduino | • PCB design, PSpice |
| • Computer-Aided Design (CAD) | • Bilingual (English/Spanish) | • Embedded Systems |

RESEARCH EXPERIENCE

Texas A&M Biomedical Engineering Young Scholars Program

Summer 2023

- Collaborated with Dr. Samuel B. Mabbott's research team at Texas A&M University to design and prototype a biomedical diagnostic device for analyzing samples used in myocardial infarction detection.

WORK EXPERIENCE

FIRST Robotics South Texas Lead Mentor

August 2024 - August 2025

- Mentored FLL, FTC, and FRC teams in robot design, electrical systems, troubleshooting, prototyping, and iterative problem-solving.
- Supported regional robotics events through team assistance, competition operations, and technical guidance for teams advancing to the FIRST World Championship.

UTRGV Industrial Training and Assessment Center (ITAC)

August 2025 - Present

- Conduct industrial energy assessments by collecting and analyzing facility electrical, equipment, and operating data to identify opportunities for energy and cost savings.
- Develop Excel-based engineering models to estimate energy consumption, demand reduction, implementation cost, annual savings, ROI, and simple payback for proposed upgrades.
- Evaluate systems and technologies including solar PV, motors, lighting, occupancy controls, compressed air, chillers, and power factor correction.
- Collaborate with faculty, engineers, facility personnel, and student researchers during site visits, technical analysis, and development of engineering recommendations.

PROJECTS

UTRGV Rocket Launchers - Rocket Telemetry Team

August 2024 - December 2024

- Designed and implemented telemetry features for a rocket launcher system, enabling data collection and monitoring to support testing, troubleshooting, and launch performance evaluation.*

LumaSense Embedded Light-Sensing System

Spring 2025

- Developed an Arduino-based light-sensing system using an LDR and 555 timer to acquire, process, and display sensor data for real-time monitoring.*

Underwater ROV Electronics System

Fall 2026

- Designed and integrated the electrical system for an underwater ROV, including camera power, control electronics, signal routing, and power distribution, while troubleshooting hardware integration and underwater system constraints.*

LEADERSHIP & PROFESSIONAL AFFILIATIONS

Society of Hispanic Professional Engineers (SHPE)

Spring 2026 - Present

Marketing Chair (Officer)

Institute of Electrical and Electronics Engineers (IEEE)

August 2024 - Present

Member